PROJECT REPORT: GARBAGE CLASSIFICATION USING DEEP LEARNING

Course Project: Automated Waste Sorting System

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Dataset Source: Kaggle (Garbage Classification Dataset)

1. Introduction & Objective

The management of solid waste is a critical environmental challenge in modern urban areas. Manual sorting of garbage is time-consuming, inefficient, and highly prone to human error. The primary objective of this project is to develop an automated solution using advanced Deep Learning techniques to classify waste into distinct streams instantly.

By building a Convolutional Neural Network (CNN) using the TensorFlow and Keras frameworks, this project automatically classifies household waste images into six targeted categories: Plastic, Glass, Paper, Cardboard, Metal, and Trash. Implementing such computer vision models in smart bins and recycling hubs can significantly optimize recycling pipelines, reduce landfill contamination, and promote environmental sustainability.

2. Dataset Exploration & Preprocessing

Data Overview

The dataset contains images belonging to 6 structural classes. A quick distribution and characteristics check was performed to analyze the dataset structure:

Waste Category	Visual Characteristics & Core Features
Cardboard	Flat or corrugated brown boxes, high texture uniformity, rigid edges.
Glass	Transparent, semi-transparent, and colored bottles or jars; reflective surfaces.
Metal	Soda cans, aluminum foil, and tin containers; metallic luster and distinct curves.
Paper	White sheets, newspapers, crumpled notes, and magazines; flat matte textures.
Plastic	Water bottles, food containers, and synthetic packaging; transparent or opaque.
Trash	Miscellaneous non-recyclable household waste, highly irregular shapes and colors.

Data Preprocessing Pipeline

Before feeding the raw images into our deep neural network, a robust preprocessing pipeline was established to standardize input data and ensure stable training:

- **Image Resizing:** All images were resized to a uniform dimension of **224 × 224 pixels**. This step is crucial because the fully connected dense layers requires static vector sizes.
- **Pixel Normalization:** Image pixels inherently range as integers from 0 to 255. We integrated a `Rescaling(1./255)` layer to linearly map these values to a floating-point scale between **[0, 1]**. This standardizes features and prevents exploding gradients.
- **Data Splitting:** To evaluate the model without data leakage or bias, the dataset was shuffled and split according to standard protocols:
  - **Training Set (80%):** Used by the network to update weights and learn essential spatial features.
  - **Validation Set (10%):** Used during the training epochs to monitor generalization and avoid overfitting.
  - **Test Set (10%):** Kept completely isolated to compute final unbiased accuracy and performance metrics.

3. Model Architecture

A custom Sequential Convolutional Neural Network (CNN) was constructed. The architecture utilizes alternating convolution and pooling layers to extract abstract spatial structures hierarchy from the waste objects.

```
Input Image (224 x 224 x 3)
      |
  [Rescaling Layer] ───────── (Normalizes pixels to 0-1 range)
      |
  [Conv2D + MaxPooling] ───── (Extracts low-level structural edges & corners)
      |
  [Conv2D + MaxPooling] ───── (Extracts mid-level shapes & contours)
      |
  [Conv2D + MaxPooling] ───── (Extracts high-level semantic object features)
      |
  [Flatten Layer] ─────────── (Converts 2D feature maps into a single 1D Vector)
      |
  [Dense Layer (128)] ─────── (Fully connected layer for feature combination)
      |
  [Dropout Layer (0.5)] ───── (Randomly deactivates 50% of neurons to prevent
memorization)
      |
  [Output Dense Layer (6)] ── (Softmax activation outputs probabilities for 6 classes)
```

The network configuration details include:

- **Optimizer:** The model was compiled using the **Adam** optimizer due to its adaptive learning rate behavior, which helps escape local minima efficiently.
- **Loss Function: Sparse Categorical Crossentropy** was selected since target labels are integers representing mutually exclusive classes.

4. Results & Discussion

Training Dynamics

The model was trained over 10 complete epochs. Throughout the training cycle, the training accuracy displayed a consistent upward trajectory while the categorical training loss dropped steadily. The validation curve closely aligned with the training metric, demonstrating that the network successfully extracted generalized shapes rather than memorizing noise.

Error Diagnosis & Confusion Matrix Insights

The generated Confusion Matrix and comprehensive Classification Report revealed valuable behavioral insights:

> **Key Discovery:** The model showed exceptional precision and F1-scores when identifying **Cardboard** and **Paper**. This is due to their consistent textures and solid matte reflections. Conversely, the primary point of confusion occurred between **Glass** and **Plastic**. This misclassification is expected because both classes exhibit transparency, high specular reflectivity, and can share virtually identical cylindrical geometric silhouettes.

5. Conclusion & Future Scope

In this project, a fully functional deep learning pipeline was built from scratch to automate waste categorization. The successful extraction of class parameters proves that computer vision is viable for commercial use in recycling sorting centers and smart city waste management frameworks.

Future Enhancements:

1. **Data Augmentation:** Implementing random real-time flips, translations, zooms, and rotations during preprocessing to expose the network to diverse angles, artificially expanding the dataset size.
2. **Transfer Learning:** Replacing the custom CNN layers with a pre-trained state-of-the-art backbone network such as **MobileNetV2** or **ResNet50**. Utilizing features learned on the massive ImageNet dataset can easily push the classification accuracy above 90%.